

5(a)	(pathogen / causative organism is) transferred / passed on / AW, (from infected) to an uninfected, person / organism ; I disease I from one person to another <i>unqualified</i> I spread	1
5(b)	any two from <i>information needed</i> 1 magnification of, image / (electron) micrograph / Fig. 5.1 ; A scale bar I magnification of (electron) microscope 2 length of flagellum (in, image / Fig. 5.1) ; I width / diameter / size 3 <i>formula</i> actual length of flagellum = measured length of flagellum divided by magnification ; A $A = I / M$ or magnification triangle	2
5(c)(i)	<u>0.2</u>	1

Question	Answer	Marks
5(c)(ii)	<i>large outbreaks of cholera in countries shown in Table 5.1 accept ora any five from:</i> 1 <i>ref. to poverty, unqualified</i> 1 civil disturbance / (civil) war ; 2 named natural disaster ; 1 famine / drought 3 lack of, uncontaminated / AW, drinking water ; A examples e.g. no piped water supply / no bottled water / no water treatment 4 <i>ref. to sewage, qualified ;</i> e.g. no sewage treatment A examples e.g. no separation of drainage from homes and water supplies / contamination of water supplies with human faeces 5 poor sanitation ; <i>credit only if mp3 or mp4 are not awarded</i> 6 <i>ref. to poor personal hygiene / described example ;</i> 7 eating contaminated foods ; e.g. crops irrigated with contaminated water / seafood harvested from contaminated seawater 8 slow(er) / no / little, diagnosis ; 9 slow(er) / no / little, treatment / healthcare / medical facilities ; 10 <u>oral</u> rehydration, (salts / therapy), not, provided / available ; A named example of oral rehydration 11 antibiotics not, provided / available ; 12 vaccine not provided / vaccine not available / reluctance to be vaccinated / ref to anti-vaccination campaigns ; 13 poor, understanding of / education about, prevention / treatment / precautions ; 14 AVP ; e.g. difficult to reach people living in, rural / remote / dangerous, areas lack of trained professionals antibiotic resistance <i>ref. to people bringing in cholera from elsewhere</i>	5